

Wazuh Security Monitoring

Purpose

Wazuh was implemented as the security-monitoring and endpoint-visibility layer of the IAM/PAM cybersecurity environment.

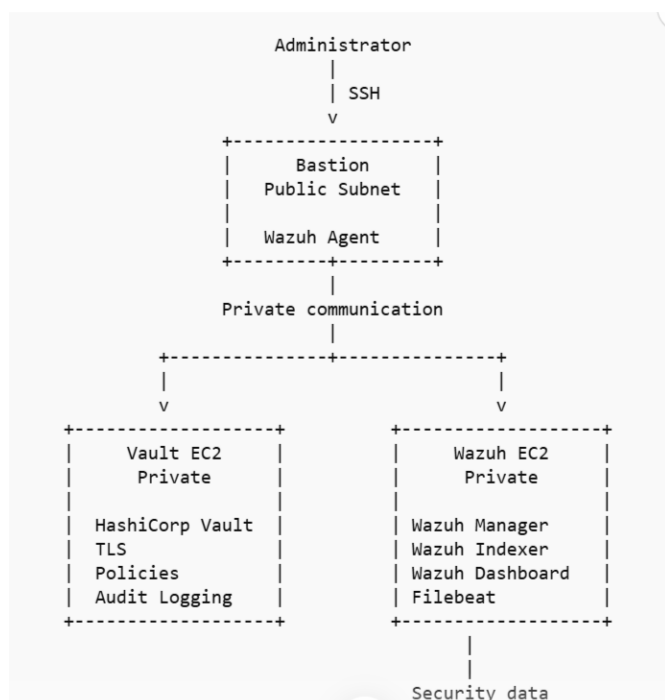
The objective was to demonstrate how a private security-monitoring platform can provide centralized visibility into activity from an administrative gateway without exposing the monitoring infrastructure directly to the Internet.

The implementation focused on:

- Private network placement
- Bastion-mediated administration
- Endpoint agent deployment
- Centralized security monitoring
- Vulnerability detection
- Security Configuration Assessment (SCA)
- File Integrity Monitoring (FIM)
- Security-event visibility
- Controlled network access
- Validation of agent-to-manager communication

Implemented Architecture

The implemented environment used a public Bastion as the controlled administrative gateway to private infrastructure.



The Bastion was not the Wazuh server.

The Wazuh platform was installed on the private EC2 instance, while the Wazuh agent was installed separately on the Bastion.

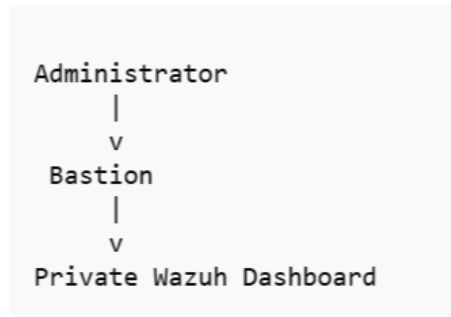
Vault remained a separate private security component and was accessed administratively through the Bastion.

Network Security Model

The Wazuh server was intentionally placed in the private subnet.

The Dashboard was not exposed through a public IP address.

Administrative access followed the pattern:



For local browser access, an SSH tunnel through the Bastion was used to reach the private Dashboard.

This preserved the private placement of the Wazuh service while allowing administrative access from the administrator's workstation.

Security-group rules restricted Wazuh communication to the Bastion security group rather than permitting unrestricted inbound access.

The Wazuh agent required communication with the Manager over the Wazuh service ports used for enrollment and agent communication. These were permitted specifically between the Bastion and the private Wazuh server.

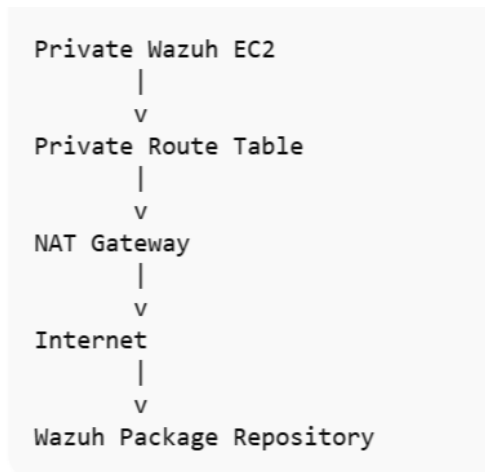
Temporary NAT Egress

The private Wazuh EC2 initially had no working Internet egress.

This was intentionally identified and validated before installation rather than assuming that a private subnet could reach external package repositories.

A temporary NAT Gateway was subsequently introduced to provide controlled outbound Internet connectivity during the Wazuh installation.

The temporary flow was:



The NAT Gateway was not used as the Wazuh log-transport path.

After Wazuh installation and validation were completed, the default route through the NAT Gateway was removed and the NAT Gateway was deleted.

The private subnet therefore returned to a model in which only the local VPC route remained.

This demonstrated an important security principle:

Temporary outbound Internet access required for installation does not need to become permanent Internet exposure.

Wazuh Components

An all-in-one Wazuh deployment was implemented on the private EC2 instance.

The deployment included:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard
- Filebeat

The four core services were validated as active after installation.

The Dashboard was also successfully reached through the private administrative path.

Bastion Agent

The Bastion was enrolled as a Wazuh endpoint.

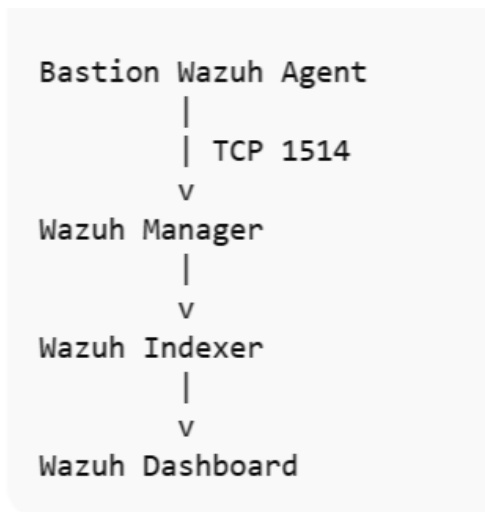
The deployed agent reported:

- Amazon Linux 2023
- Wazuh 4.12.0
- Active endpoint status
- Successful communication with the Wazuh Manager

Agent-to-manager communication was explicitly validated over TCP 1514.

Agent enrollment required communication over TCP 1515 and was also validated after the appropriate private security-group rule was added.

This provided an end-to-end validation path:



Security Monitoring Capabilities

The deployed Wazuh environment provided several security-monitoring capabilities.

Vulnerability Detection

The Bastion endpoint produced vulnerability findings during validation.

Observed results included:

- 6 High
- 11 Medium
- 1 Low

These findings demonstrated that the endpoint was being evaluated for vulnerable installed software rather than merely appearing as a registered Wazuh agent.

Security Configuration Assessment

Wazuh performed a CIS-oriented Security Configuration Assessment against the Amazon Linux 2023 endpoint.

The observed validation included:

- 75 passed
- 102 failed
- 6 not applicable
- Overall score: 42%

The result provided useful evidence that security monitoring was evaluating configuration posture in addition to collecting endpoint telemetry.

The score was treated as a validation finding rather than as a claim that the endpoint was fully hardened.

File Integrity Monitoring

FIM was enabled and configured for monitored system paths including:

- /etc
- /usr/bin
- /usr/sbin
- /bin
- /sbin
- /boot

A controlled file-change test was attempted during validation.

The test did not produce a corresponding FIM event in the Dashboard during the short observation window.

Therefore, this project does **not** claim successful end-to-end FIM event detection.

The result was documented as a validation limitation rather than being represented as a successful test.

Agent connectivity and the other Wazuh monitoring functions were independently demonstrated.

Validation Summary

The following implementation results were established:

Control / Capability	Result
Wazuh installation	Successful
Wazuh Manager	Active
Wazuh Indexer	Active
Wazuh Dashboard	Active
Filebeat	Active
Private Dashboard access	Validated
Bastion agent enrollment	Successful
Agent status	Active
Agent → Manager communication	Validated
Vulnerability detection	Validated
SCA	Validated
FIM configuration	Validated
FIM event detection test	Attempted; event not observed
Public Wazuh endpoint	None

Control / Capability	Result
Temporary NAT egress	Used during installation
NAT removal	Completed

Security Controls Demonstrated

The Wazuh implementation reinforced several controls used throughout the wider project:

Network Segmentation

Security-monitoring infrastructure was placed in a private subnet rather than directly exposing the Dashboard to the Internet.

Bastion-Mediated Administration

Administrative access to private infrastructure was performed through the Bastion.

Security-Group Restriction

Wazuh communication was restricted to the Bastion security group rather than broadly permitting external sources.

Temporary Egress

Internet access required for package installation was provided temporarily through NAT and removed after installation.

Centralized Monitoring

The Bastion endpoint reported security telemetry to a centralized private Wazuh Manager.

Independent Security Evidence

Wazuh provided security-monitoring evidence that complemented AWS-level audit information such as CloudTrail.

Relationship to IAM and PAM

Wazuh does not replace IAM or PAM.

The controls address different questions:

IAM

Who are you?

What are you authorized to access?

PAM

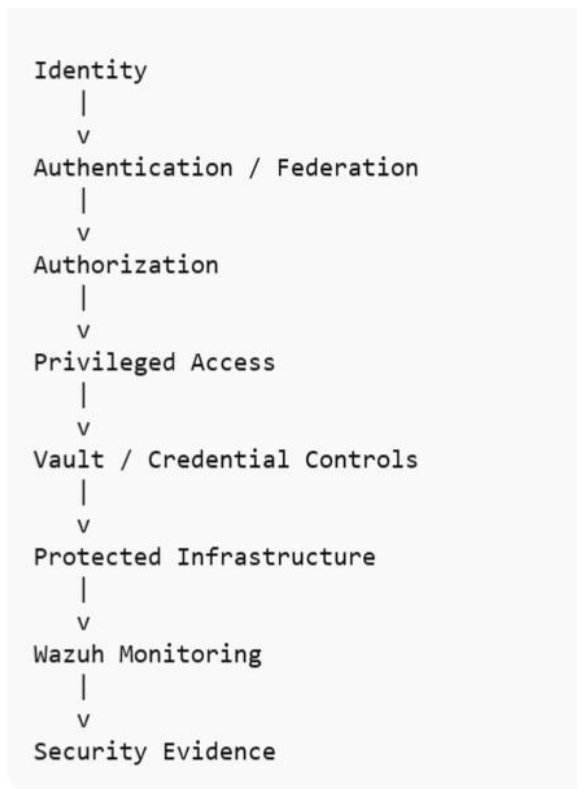
How is privileged access requested,
approved, controlled, limited, and expired?

Wazuh

What security telemetry and endpoint activity

can we monitor and investigate?

In this project, the layers work together conceptually:



This creates a broader security-control model rather than treating identity, privilege, secrets, and monitoring as isolated technologies.

Lessons Learned

Private does not mean unreachable

A private EC2 instance can remain inaccessible from the Internet while still being administratively reachable through a controlled Bastion path.

Installation requirements can temporarily differ from steady-state requirements

Wazuh installation required outbound package access, but ongoing Dashboard administration did not require permanent Internet exposure.

The temporary NAT architecture therefore served an installation purpose rather than becoming part of the permanent monitoring design.

Connectivity problems should be isolated by layer

The Wazuh agent deployment exposed two separate network requirements:

- TCP 1515 for enrollment
- TCP 1514 for agent communication

Troubleshooting these independently helped distinguish security-group restrictions from general Internet-egress problems.

Security findings are useful evidence

The SCA and vulnerability results demonstrated that the platform was doing more than simply running its services.

Failed validation is still valuable

The FIM test did not produce the expected Dashboard event during the validation window.

Rather than claiming success, the result was documented as a limitation.

This reinforces the broader project methodology:

Security controls should be tested against observable outcomes, and inconclusive results should remain documented as inconclusive.

Portfolio Scope

This document describes the implemented Wazuh environment at an architectural and validation level.

It intentionally excludes:

- Credentials
- API keys
- Private keys
- Passwords
- Private IP addresses
- AWS account identifiers
- Instance identifiers
- Internal hostnames
- Authentication links
- Production data
- Complete deployment commands
- Turnkey reproduction instructions

The purpose of the public portfolio is to demonstrate architecture, security reasoning, implementation experience, validation methodology, troubleshooting, and lessons learned rather than provide a reproduction guide for the original environment.